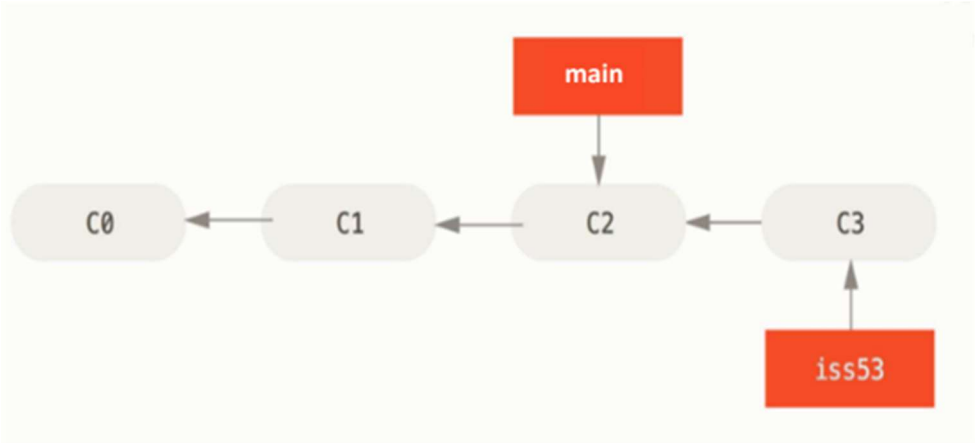
1) Explain the purpose and objectives of each of the five phases in the DevOps lifecycle.

2) Analyze the benefits and challenges of implementing Continuous Integration (CI) and Continuous Delivery (CD) in a large-scale software project. How do these practices impact the team's productivity, code quality, and time-to-market?

3) Utilizing knowledge of version control systems, write down the specific commands required to execute the described changes.

   i. Add an h1 tag to 'README.md' and commit the changes (C0)
   ii. Create a new txt file and commit the creation (C1)
   iii. Modify the txt file and commit the changes (C2)
   iv. Create a new branch named "issue53"
   v. Create the "index.html" file and update "readme.md" in the "issue53" branch, then commit the changes (C3) and push it to GitHub.



4) Write a shell script that continuously monitors the system uptime and stores the data in a TSV file. The script should run indefinitely, updating the uptime data every 5 minutes. Arrange the entries in the file so that the latest entry comes first, based on timestamp order.

5) Differentiate between RPM-based and Debian-based package management systems.

6) Evaluate the effectiveness of the Maven Build Lifecycle in managing the build process for Java projects.

7) State the purpose of the hotfix branch and elaborate on the git status, git add, and git config commands.

8) Write a configuration settings required in a Vagrantfile to provision a CentOS Stream 9 virtual machine for hosting a simple web app. Include details such as network settings and provisioning scripts.

9) Describe the steps required to connect two Docker containers using a custom Docker network.

10) Write a configuration settings required in a Vagrantfile to provision an Ubuntu Jammy 64-bit virtual machine for hosting a simple web app. Include details such as network settings, synced folders, and provisioning scripts.

11) Describe the step-by-step process of containerizing an HTTP Node server. Include the necessary commands and configuration settings, as well as an explanation of each step's purpose.

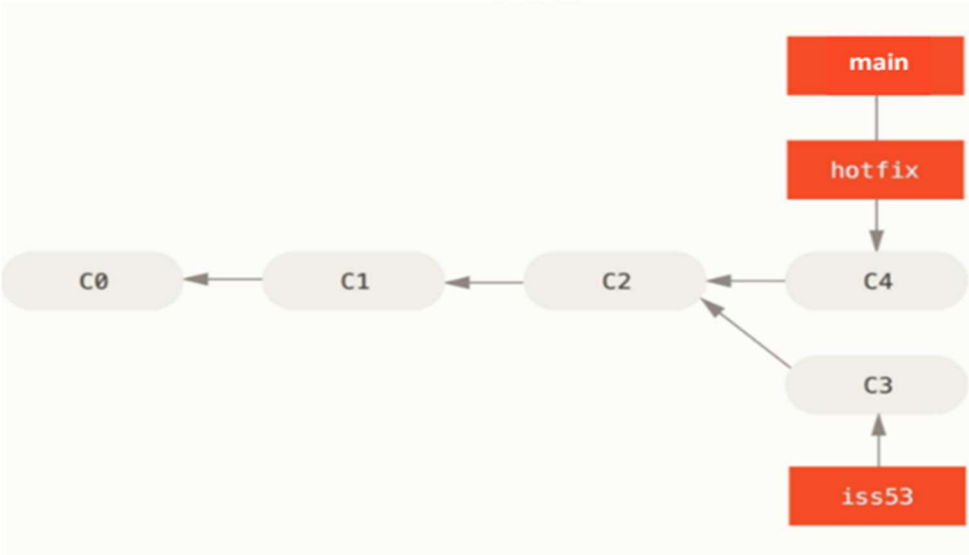
12) Describe how the “git log”, “git checkout”, and “git push -u” commands are used in the context of branching. Provide step-by-step explanations of each command's functionality and typical usage scenarios.

13) Analyze the sequence of phases in the Maven build lifecycle, particularly focusing on the verify, install, and deploy phases with.

14) Distinguish between the waterfall and agile software development models.

15) Using knowledge of Git, write down the specific commands required to execute the described actions.

i. Difference between iss53 and hotfix
ii. Difference between commit C1 & C2
iii. Merge the 'hotfix' branch into the 'main' branch and delete the 'hotfix' branch.
iv. Merge the 'iss53' branch into the 'main' branch and delete the 'iss53' branch



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16) Describe the DevOps tools used for source control and collaborative coding, as well as continuous delivery, and provide examples of such tools.

17) Explain the functionality and significance of the commands in a Unix/Linux environment:

i. uptime
ii. cat /proc/uptime
iii. free -m
iv. rm -rf *
v. df -h

18) Describe the specific functionalities of the /etc, /mnt, and /tmp directories. Provide examples of scenarios where each directory might be utilized in system administration tasks.

CHAPTER 1

1) Describe any five phases in the DevOps lifecycle.
2) Explain the process of code build and deployment to production within the DevOps lifecycle, along with examples of tools commonly utilized in each stage.
3) Explain a scenario involving continuous integration (CI) and continuous delivery (CD).
4) Describe the DevOps tools used for source control and collaborative coding, as well as continuous delivery, and provide examples of such tools.
5) Distinguish between the waterfall and agile software development models.
6) Write a short note on SDLC.

CHAPTER 2

1) Differentiate between RPM-based and Debian-based package management systems.
2) State any two Linux principles and elaborate on any two important directories in Linux.
3) Write a shell script that continuously monitors the system uptime and stores the data in a TSV (Tab Separated Values) file. The script should run indefinitely, updating the uptime data every 5 minutes. Ensure that the entries in the file are arranged in descending order of timestamp.
4) Write a shell script that continuously monitors system uptime, memory utilization, and disk utilization, storing the data in a TSV file. The script should run indefinitely, updating the uptime data every 5 minutes, and ensuring that entries in the file are arranged in descending order of timestamp.
5) Explain Linux principles and state any three advantages of Linux.
6) State the purpose of the following commands:
   a. Uptime
   b. cat /proc/uptime

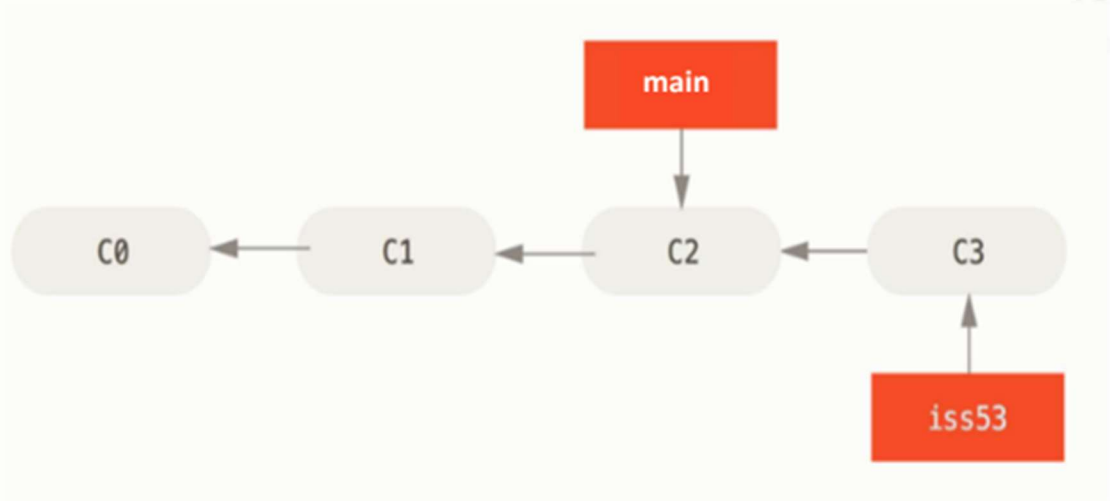
c. free –m

d. rm -rf *

e. df –h

CHAPTER 3

1) Write down the commands corresponding to the following diagram:



a. Add an h1 tag to 'README.md' and commit the changes (C0)

b. Create a new txt file and commit the creation (C1)

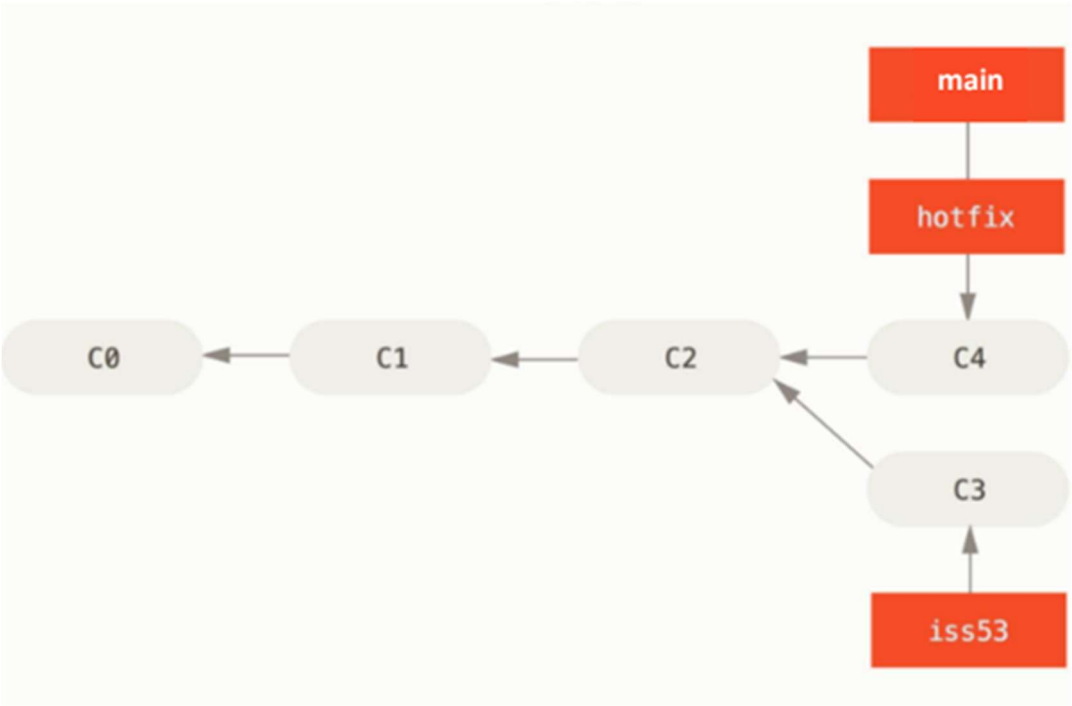
c. Modify the txt file and commit the changes (C2)

d. Create a new branch named "issue53"

e. Create the "index.html" file and update "readme.md" in the "issue53" branch, then commit the changes (C3) and push it to GitHub.

2) State the purpose of the hotfix branch and elaborate on the git status, git add, and git commit commands.

3) Write down the commands corresponding to the following diagram:



a. Show difference between iss53 and hotfix
b. Show difference between commit C1 & C2
c. Merge the 'hotfix' branch into the 'main' branch and delete the 'hotfix' branch.
d. Merge the 'iss53' branch into the 'main' branch and delete the 'iss53' branch.

4) State branching in Git and elaborate on the git log, git checkout, and git push -u origin main commands.
5) Describe the validate, compile, test, package, verify, and install phases in Maven Build Lifecycle.
6) State any two examples of build tools and elaborate verify, install, and deploy phases in Maven.
7) Describe how to containerize an HTTP Node server using Docker, including the necessary steps and a Dockerfile example.
8) Explain the steps involved in pushing an image to Docker Hub.
9) Describe the steps required to connect two Docker containers using a custom Docker network.
10) Write a short note on Amazon Elastic Container Services.
11) Describe the Vagrantfile configuration to provision a simple web app on a CentOS Stream 9 virtual machine.
12) Describe the Vagrantfile configuration to provision a simple web app on an Ubuntu Jammy 64-bit virtual machine.